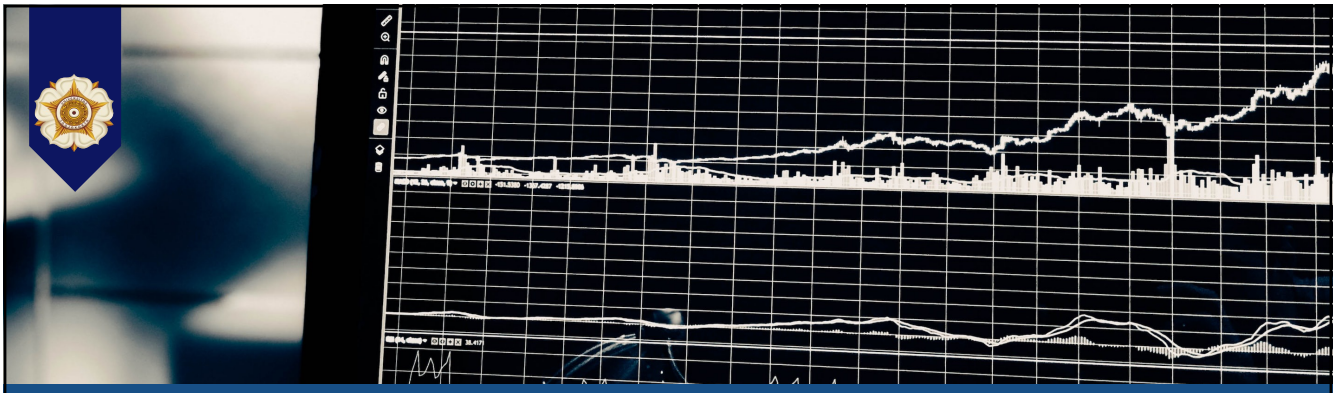


Introduction to Statistics

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Department of Electrical and Information Engineering
Faculty of Engineering
Universitas Gadjah Mada
INDONESIA

Version: February 2025

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1

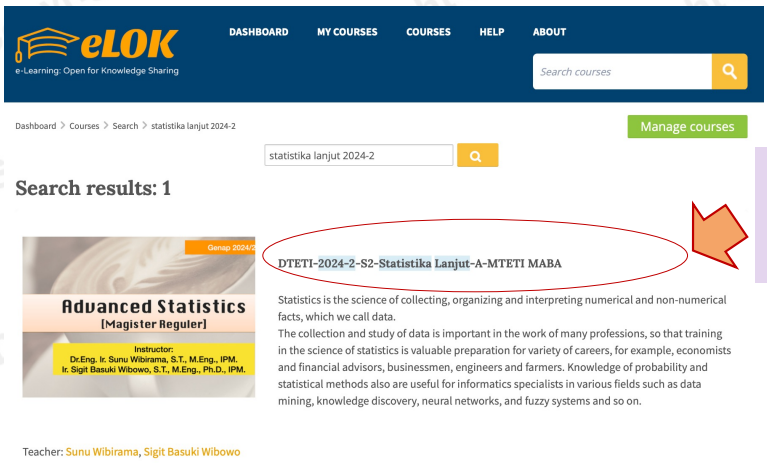
About this course

- We will learn about concept of statistics and its implementation in some study cases (related with postgraduate research).
- Mas Kristian **Adi** Nugraha, S.Kom, M.T. will be our teaching assistant (email: kristianadinugraha@mail.ugm.ac.id).
- We need a contact person in your class as a representative for academic matters (please discuss and pass the information to Mas Adi : Name, NIM, and Whats App contact).

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e-Learning page (Class A / Reguler Master Program)



Pay attention to the course's name

https://elok.ugm.ac.id/course/view.php?id=15913
Enrollment key: STATISTIKA2025

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Schedule (Class A / Magister Reguler)

Kamis, 13:00 - 15:30 Ruang 10B3	1	Kamis 13-02-2025 13:00-15:30	10B3	Dr.Eng. Ir. Sunu Wibirama, S.T., M.Eng., IPM.	TEE226101	Statistika Lanjut	2.00	MABA	40
	2	Kamis 20-02-2025 13:00-15:30	10B3	Dr.Eng. Ir. Sunu Wibirama, S.T., M.Eng., IPM.					
	3	Kamis 27-02-2025 13:00-15:30	10B3	Dr.Eng. Ir. Sunu Wibirama, S.T., M.Eng., IPM.					
	4	Kamis 06-03-2025 13:00-15:30	10B3	Dr.Eng. Ir. Sunu Wibirama, S.T., M.Eng., IPM.					
	5	Kamis 13-03-2025 13:00-15:30	10B3	Dr.Eng. Ir. Sunu Wibirama, S.T., M.Eng., IPM.					
	6	Kamis 20-03-2025 13:00-15:30	10B3	Dr.Eng. Ir. Sunu Wibirama, S.T., M.Eng., IPM.					
	7	Kamis 27-03-2025 13:00-15:30	10B3	Dr.Eng. Ir. Sunu Wibirama, S.T., M.Eng., IPM.					
	8	Kamis 03-04-2025 13:00-15:30	10B3	Dr.Eng. Ir. Sunu Wibirama, S.T., M.Eng., IPM.					
	9	Kamis 24-04-2025 13:00-15:30	10B3	Ir. Sigit Basuki Wibowo, S.T., M.Eng., Ph.D., IPM.					
	10	Kamis 01-05-2025 13:00-15:30	10B3	Ir. Sigit Basuki Wibowo, S.T., M.Eng., Ph.D., IPM.					
	11	Kamis 08-05-2025 13:00-15:30	10B3	Ir. Sigit Basuki Wibowo, S.T., M.Eng., Ph.D., IPM.					
	12	Kamis 15-05-2025 13:00-15:30	10B3	Ir. Sigit Basuki Wibowo, S.T., M.Eng., Ph.D., IPM.					
	13	Kamis 22-05-2025 13:00-15:30	10B3	Ir. Sigit Basuki Wibowo, S.T., M.Eng., Ph.D., IPM.					
	14	Kamis 29-05-2025 13:00-15:30	10B3	Ir. Sigit Basuki Wibowo, S.T., M.Eng., Ph.D., IPM.					
	15	Kamis 05-06-2025 13:00-15:30	10B3	Ir. Sigit Basuki Wibowo, S.T., M.Eng., Ph.D., IPM.					

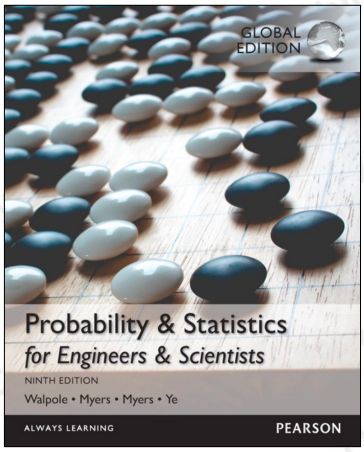
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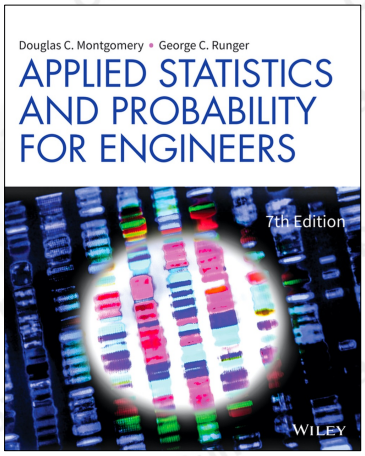
8x (including midterm exam)

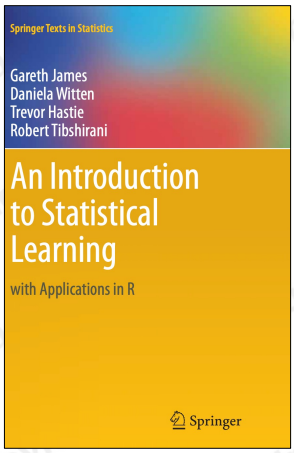
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Main reading materials



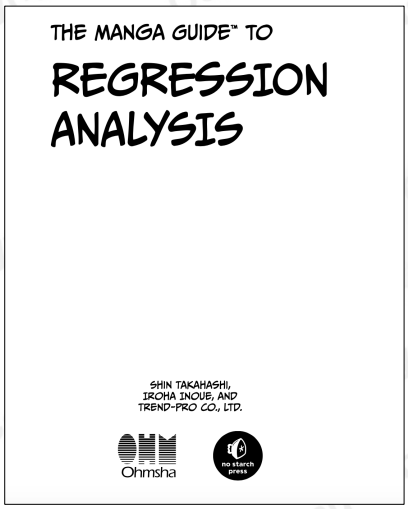


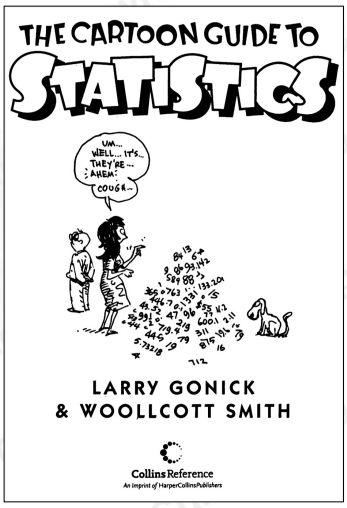


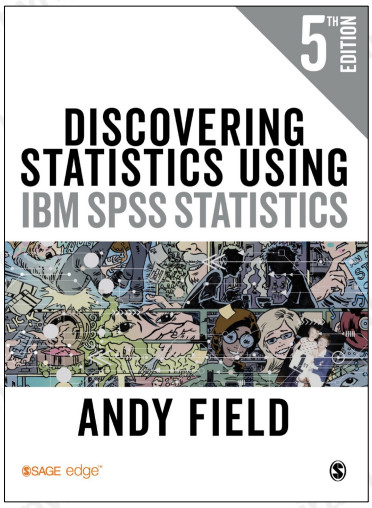
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Supporting reading materials







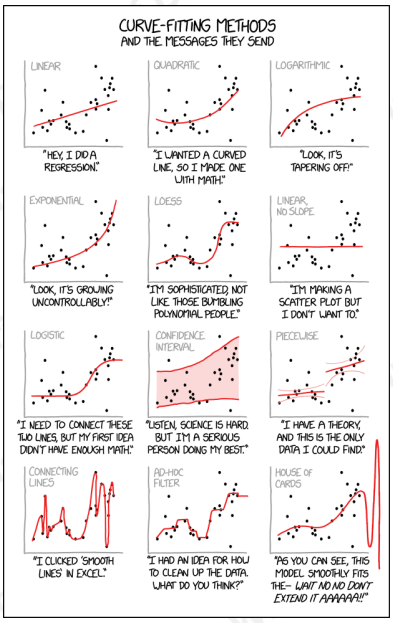
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Overview of today's lecture

Review of basic concepts in Statistics

- Statistics and Its Implementation
- Variability in Statistics
- Understanding Data and How to Get Them
- Descriptive Statistics and Data Visualization



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Part 1

STATISTICS AND ITS IMPLEMENTATION

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What is statistics?

- The field of statistics deals with the collection, presentation, analysis, and use of data to:
  - Make decisions
  - Solve problems
  - Design products and processes

Statistics is the science of learning information from data.



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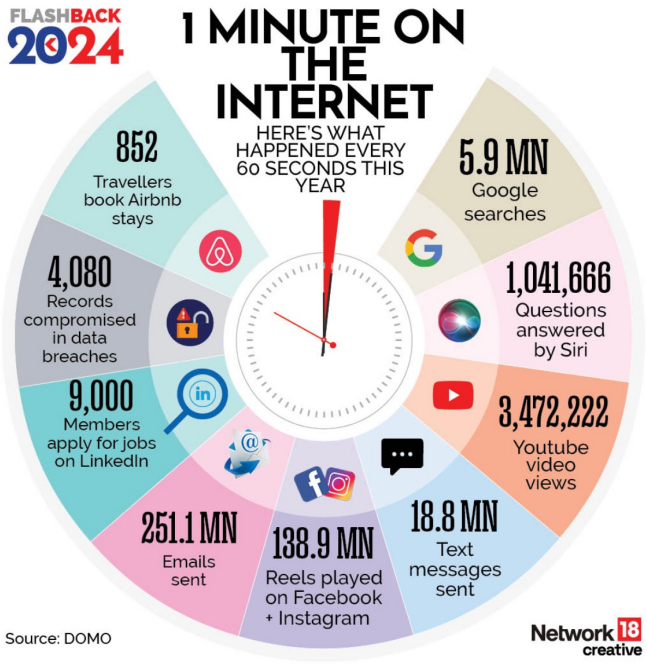
Why statistics is important?

- We need tool to model uncertainty in data → practically useful in big data analysis
- Statistics are the core ingredients of the most emerging technology today: Artificial Intelligence.
- Statistics is a field studied outside Engineering as well → you can use it as "common language across different disciplines" (example: Applied Psychology, Business and Management, Economy, Medical and Health Science, Pharmacy, Agrotechnology, etc.).



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The biggest impact of Industry 4.0:

we are great on producing data,
but struggling to understand the underlying human behaviour

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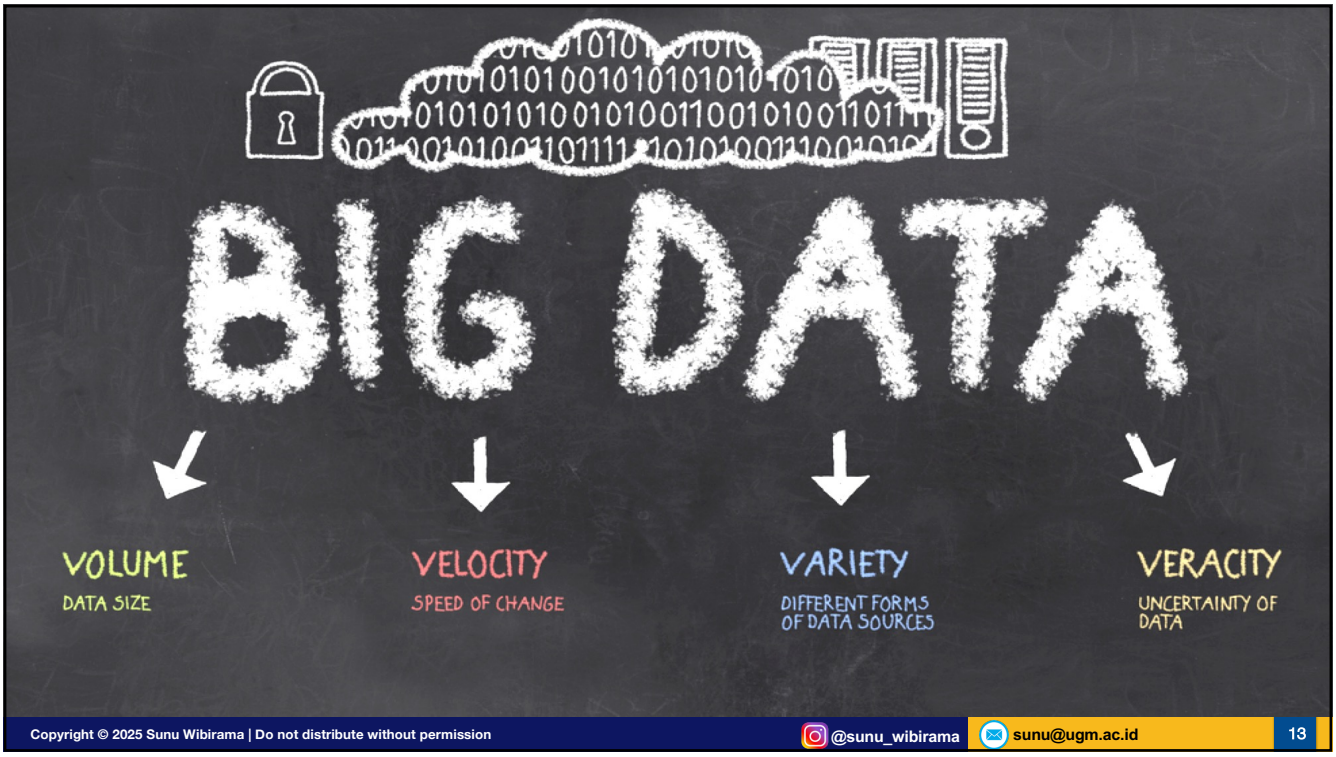
What is Big Data?

- Several other ways to think about how “Big” is “Big Data”:
  - When the size of data is a challenge.
    Data without limit of size.
  - When the data can’t be fit on one machine
  - When the data is part of cultural and daily activities
  - Collecting and using a lot of data rather than small samples
  - Accepting messiness in our data



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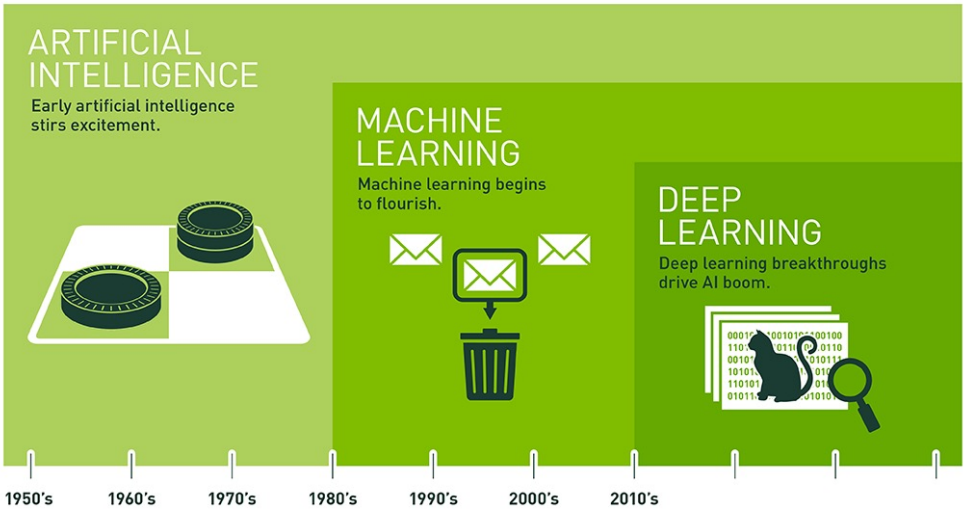
What we should know about data?

- Type of data
- Quality of data
- Preprocessing of data
- Relationship of data

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Statistics and artificial intelligence

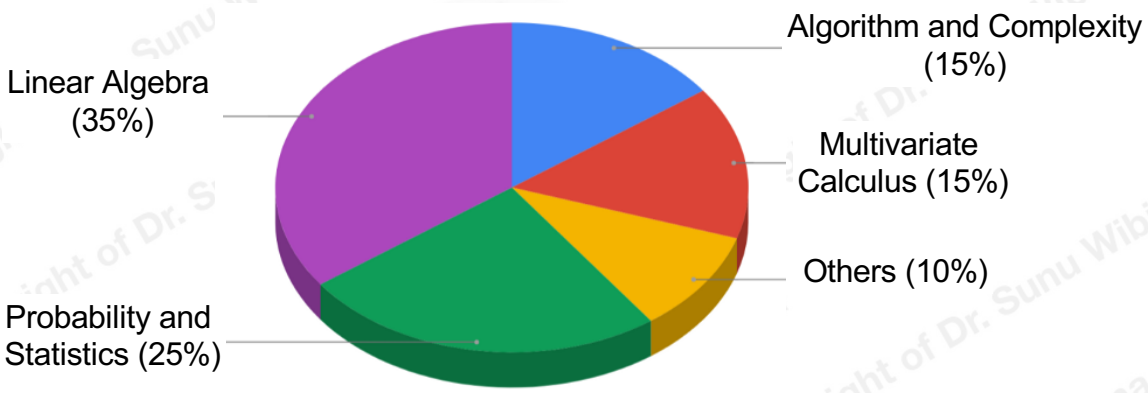


Since an early flush of optimism in the 1950s, smaller subsets of artificial intelligence – first machine learning, then deep learning, a subset of machine learning – have created ever larger disruptions.

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Statistics and artificial intelligence

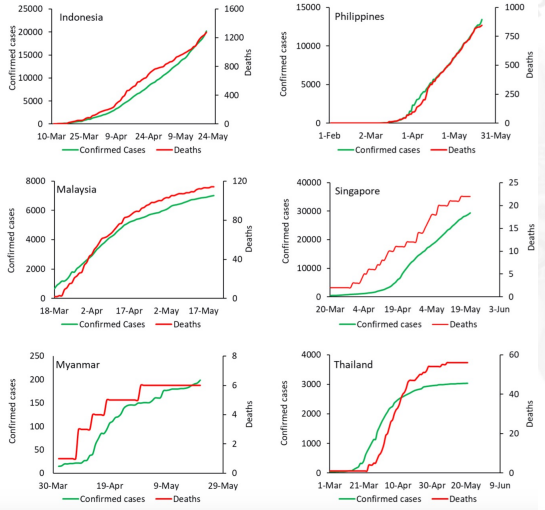


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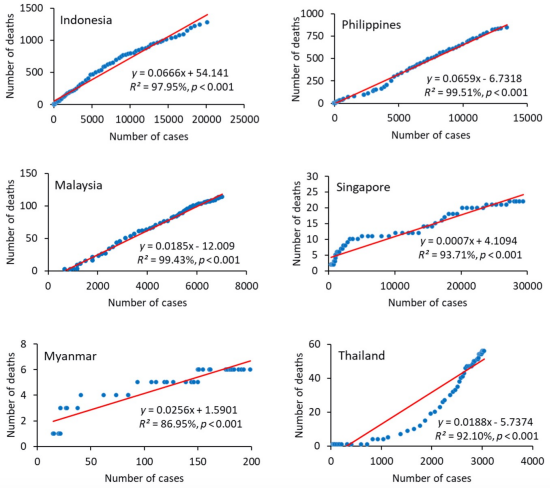
16

Statistics for modeling Covid-19 infection

Deaths and confirmed cases of infections of COVID-19 among the SEA countries as of May 21, 2020



Scatterplots and regression lines of COVID-19 confirmed cases of infections and deaths of the six SEA countries as of May 21, 2020



Puno, G.R., Puno, R.C.C. and Maghuyop, I.V., 2021. COVID-19 case fatality rates across Southeast Asian countries (SEA): a preliminary estimate using a simple linear regression model. *Journal of Health Research*.

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Part 2

VARIABILITY IN STATISTICS

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What do engineers do?

An engineer is someone who solves problems of interest to society with the efficient application of scientific principles by:

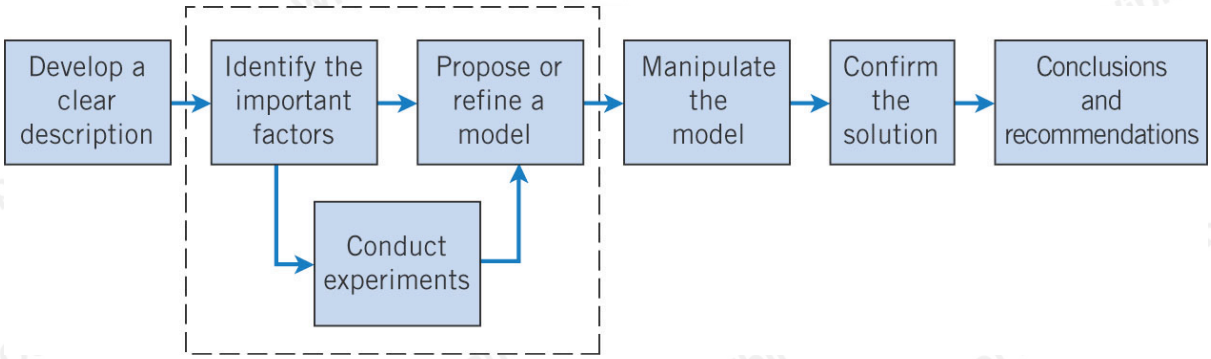
- Refining existing products
- Designing new products or processes
- Evaluating the performance of the products



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The creative process of engineering



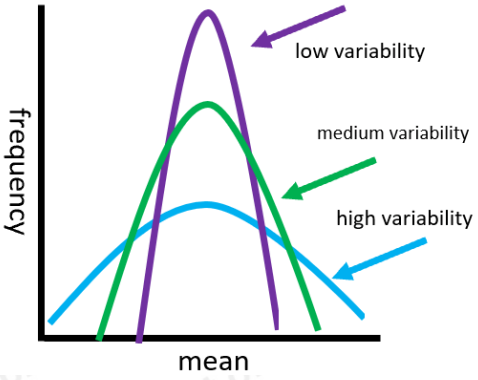
Montgomery DC, Runger GC. Applied statistics and probability for engineers. John wiley & sons; 2010 Mar 22.

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Understanding variability

- Statistical techniques are useful for describing and understanding **variability.**
- By variability, we mean successive observations of a system or phenomenon **do not produce exactly the same result.**
- Statistics gives us a framework for describing this variability and for learning about potential **sources of variability.**



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An engineering example of variability

- An engineer is designing a nylon connector to be used in an automotive engine application.
- The engineer is considering establishing the design specification on wall thickness at 3/32 inch.
- However, he is somewhat uncertain about the effect of this decision on the connector pull-off force.
- If the pull-off force is too low, the connector may fail when it is installed in an engine. Eight prototype units are produced and their pull-off forces measured (in pounds):

12.6, 12.9, 13.4, 12.3, 13.6, 13.5, 12.6, 13.1



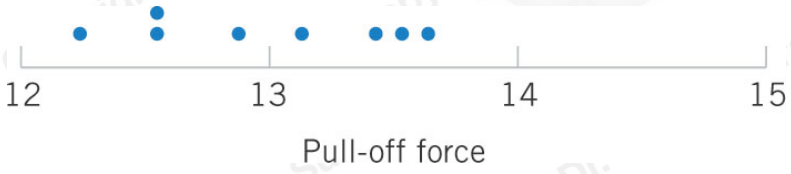
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An engineering example of variability

The dot diagram is a very useful plot for displaying a small body of data - say up to about **20 observations**.

This plot allows us to see easily two features of the data; the **location** and the **scatter or variability**.





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Variability of pull-off forces

- Not all of the prototypes have the same pull-off force. There is variability in pull-off force measurements
- Since pull-off force varies or exhibits variability, it is a **random variable**.
- A random variable, X, can be modeled by:

$$X = \mu + \varepsilon$$

where μ is a constant and ε is a random disturbance.



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An engineering example of variability

- The engineer considers an alternate design and eight prototypes are built and pull-off force measured.
- The dot diagram can be used to compare two sets of data.





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Some questions about the data

- Is it adequate to represent result of all prototypes by using only 8 samples? → random sampling and estimation
- How do we know that another sample of prototype will not give us different results? → inferential statistics can answer
- If we use current data (increasing thickness will increase the strength), what risks are associated with this decision? → data-driven business decision



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Part 3

UNDERSTANDING DATA AND HOW TO GET THEM

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Data and sampling errors

- Data from all population: census
- Data from selected individuals of population: sample
- Statistical inference can result in errors (sampling errors). This can be minimized by:
  - Determining appropriate **sample size**
  - Taking sample from population appropriately **(sampling method)**



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Basic types of studies

Three basic methods for collecting data:

- A **retrospective** study using historical data:
  Data collected in the past for other purposes.
- An **observational** study:
  Data, presently collected, by a passive observer.
- A **designed experiment:**
  Data collected in response to process input changes. The observer changes variable of system to see the results.



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Types of data

- To analyze data, we must understand type of data:
  - What you can do and can't do with different types of data
  - Helping you transforming data from one type to other types
- Four types of data:
  - Nominal data
  - Ordinal data
  - Interval data
  - Ratio data



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Nominal data

- Unordered group or categories
- Nominal values provide only enough information to distinguish one object from another. Without order, cannot say one is better than another
- Examples:
  - Zip codes
  - Employee ID numbers
  - Eye color
  - Types of operating systems



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Ordinal data

- Ordered group or categories
- Data are ordered in certain ways but intervals between measurements are not meaningful
- Examples:
  - Self-reported data / questionnaire
  - Website rating (excellent, good, fair, poor)

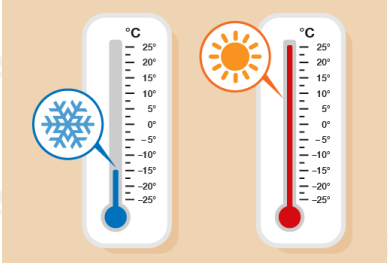


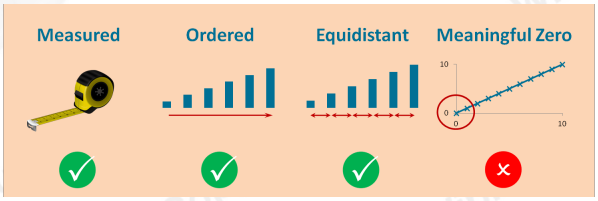
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Interval data

- Continuous data where differences between measurements are meaningful
- Zero point on the scale is arbitrary (i.e. has not absolute meaning)
- Examples:
  - calendar dates
  - temperature in Celsius or Fahrenheit





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Ratio data

- Same with interval data, but with meaningful zero
- Examples:
  - Difference between a person of 38 and 35 years old is same as difference between a person of 15 and 12 years old
  - Time to complete a particular task
  - Distance, length, weight, height



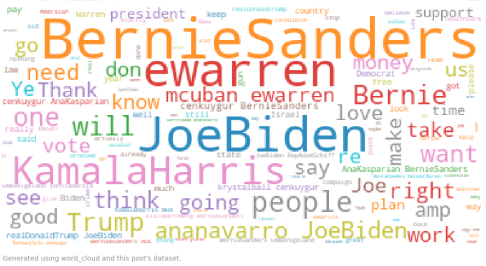


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Other classifications of data

- **Binary data:** A set of just two values (e.g., gender)
- **Textual data:** free form, usually short text data (e.g.: name, address)
- **Numeric data:** True numeric values that allow arithmetic operations (e.g., price, age)





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Data are messy

- Real world data are messy, 80% effort for most data scientist / statistican / AI engineers is to clean and pre-process the data.
- Garbage in – garbage out : do not throw your data right away to your statistical analysis model, because you will get “wrong” results.
- Data is not reality:
  - Humans define phenomenon that they want to measure
  - They design system to collect data
  - They clean and pre-process the data
  - They interpret the results

  Even with the same datasets, two people can form vastly different conclusions



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37

Noisy dataset

color	director_name	duration	gross	movie_title	anguage	country	budget	title_year	imdb_score
Color	Martin Scorsese	240	116866727	The Wolf of Wall StreetÂ	English	USA	100000000	2013	8.2
Color	Shane Black	195	408992272	Iron Man 3Â	English	USA	200000000	2013	7.2
color	Quentin Tarantino	187	54116191	The Hateful EightÂ	English	USA	44000000	2015	7.9
Color	Kenneth Lonergan	186	46495	MargaretÂ	English	usa	14000000	2011	6.5
Color	Peter Jackson	186	258355354	The Hobbit: The Desolation of SmaugÂ	English	USA	225000000	2013	7.9
	N/A	183	330249062	Batman v Superman: Dawn of JusticeÂ	English	USA	250000000	202	6.9
Color	Peter Jackson	-50	303001229	The Hobbit: An Unexpected JourneyÂ	English	USA	180000000	2012	7.9
Color	Edward Hall	180		RestlessÂ	English	UK		2012	7.2
Color	Joss Whedon	173	623279547	The AvengersÂ	English	USA	220000000	2012	8.1
Color	Joss Whedon	173	623279547	The AvengersÂ	English	USA	220000000	2012	8.1
	Tom Tykwer	172	27098580	Cloud AtlasÂ	English	Germany	102000000	2012	-7.5
Color	Null	158	102515793	The Girl with the Dragon TattooÂ	English	USA	90000000	2011	7.8
Color	Christopher Spencer	170	59696176	Son of GodÂ	English	USA	22000000	2014	5.6
Color	Peter Jackson	164	255108370	The Hobbit: The Battle of the Five ArmiesÂ	English	New Zealand	250000000	2014	7.5
Color	Tom Hooper	158	148775460	Les MisÃ©rablesÂ	English	USA	61000000	2012	7.6
Color	Tom Hooper	158	148775460	Les MisÃ©rablesÂ	English	USA	61000000	2012	7.6

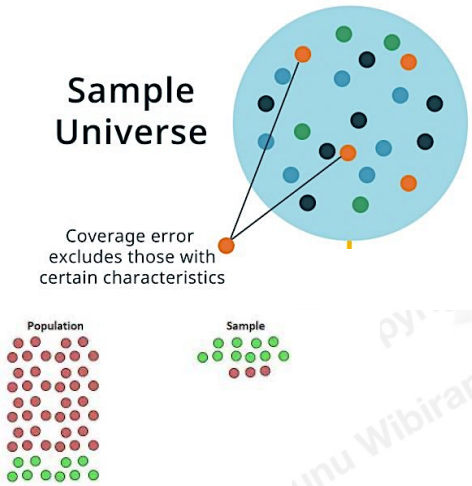
Source: https://medium.com/well-red/cleaning-a-messy-dataset-using-python-7d7ab0bf199b

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sunu@ugm.ac.id
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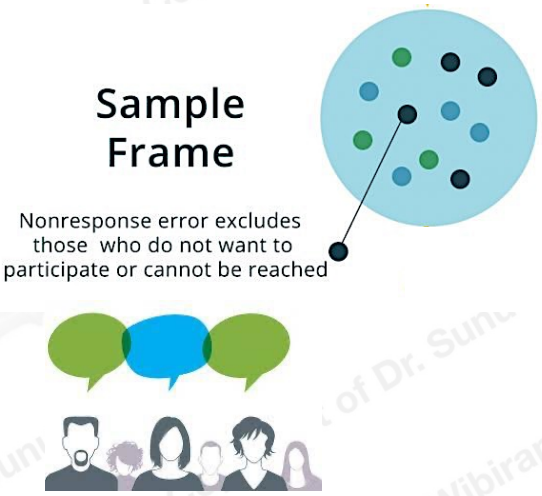
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Sampling error

Coverage error



Nonresponse error



[Source: https://www.geopoll.com/blog/sample-frame-sample-error-research]

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Part 4

DESCRIPTIVE STATISTICS AND DATA VISUALIZATION

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Descriptive vs Inferential Statistics

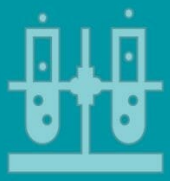
Descriptive and inferential statistics are two main branches of statistics that are used to analyze and interpret data.

DESCRIPTIVE

Descriptive statistics is a branch of statistics used to summarize and describe the characteristics of a dataset. Descriptive statistics involves calculating summary measures, such as the mean, median, mode, range, standard deviation, variance.



VS



INFERENTIAL

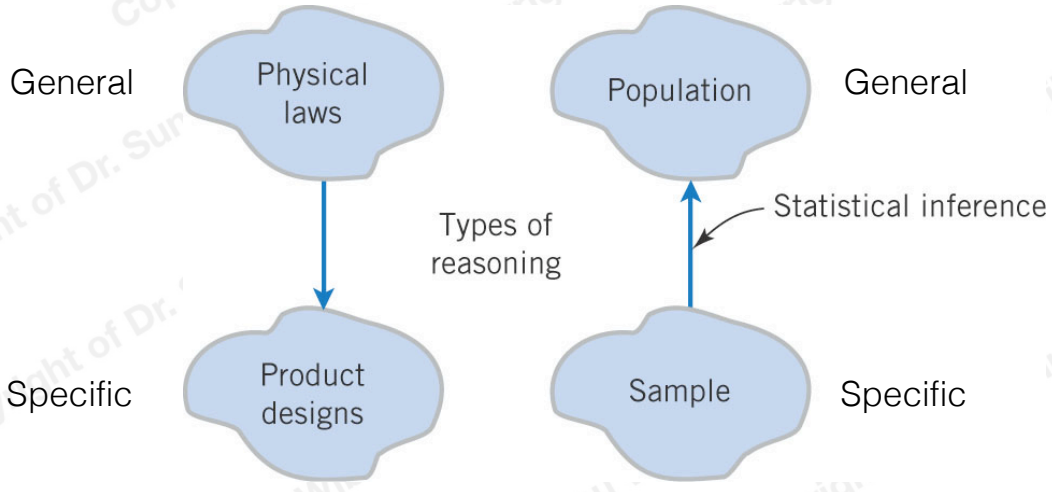
Inferential statistics is a branch of statistics used to make inferences or predictions about a population based on a sample of data. Inferential statistics involves using statistical tests, such as hypothesis tests and regression analysis.

https://www.qualitygurus.com/inferential-and-descriptive-stats-whats-the-difference/

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Two directions of reasoning

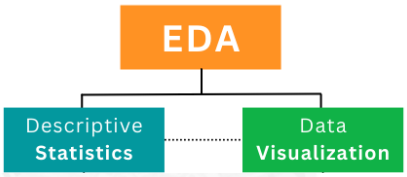


Statistical inference is one type of reasoning.

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42

Two approaches of exploratory data analysis



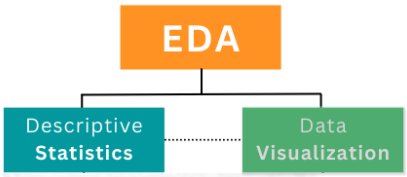
- **Non-graphical approaches** summarize and describe the dataset in a numerical way.
- There are statistical measures such as mean, median, skewness, and standard deviation to understand the central tendency, variability, and distribution of the data.

- **Graphical approaches** visually represent the data set and help identify patterns, relationships, and anomalies in the data.
- Graphical methods such as histograms, scatter plots, and box plots can reveal trends and deviations in the data that might not be immediately apparent from the numerical summaries.

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Two approaches of exploratory data analysis



- **Non-graphical approaches** summarize and describe the dataset in a numerical way.
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- **Graphical approaches** visually represent the data set and help identify patterns, relationships, and anomalies in the data.
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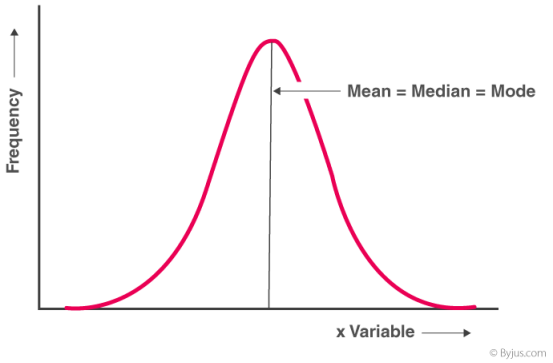
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44

Measures of central tendency

- Mean (average of all data)
- Median (middle value of the data)
- Mode (the number that occurs most often in your data)

Central tendency is important to summarize the data, so you know the basic properties of the data



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Mean

Sample mean of x_1, x_2, …, x_n from *n* observations is denoted by $\overline{x}$

$$\overline{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \sum_{i=1}^{n} \frac{x_i}{n}$$

Population mean from *N observations* can be computed by

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$$

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Median

$$\tilde{x} = \begin{cases} x_{(n+1)/2} & \text{if n is odd} \\ \frac{1}{2}\left(x_{n/2} + x_{(n/2)+1}\right) & \text{if n is even} \end{cases}$$

Example:
The median of 7, 4, 3, 5, 6, 8, and 10 is 6
[3, 4, 5, **6**, 7, 8, 10]

The median of 7, 4, 3, 5, 6, 8, 10, 1 is 5.5
[1, 3, 4, **5**, **6**, 7, 8, 10]

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47

Mode

Value *x* that occurs most often in your data

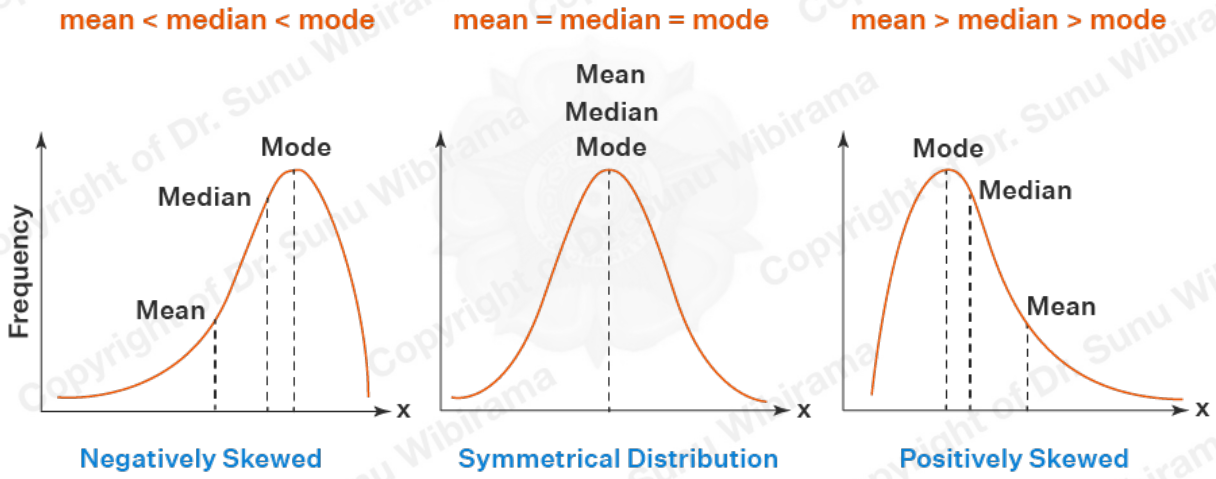
Can you find the mode from this data?

70	88	95	101	106
79	93	96	101	107
83	93	97	103	108
86	93	97	103	112
87	95	98	106	115

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Understanding distribution of the data



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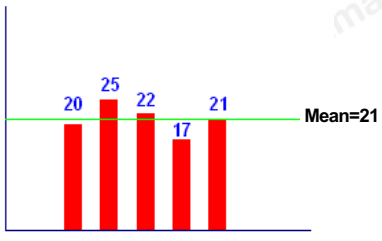
49

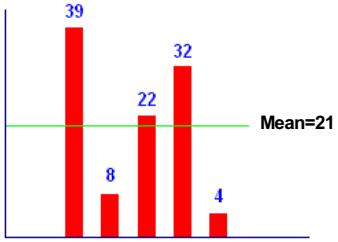
Range and variance of the data

- Range of quantitative data is the difference between maximum and minimum value

  Range = Maximum – Minimum

- Variance measures how far the data are spread from the average value (mean)





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Variance and standard deviation of population

Variance of population

$$\sigma^2 = \sum_{i=1}^{N} \frac{(x_i - \mu)^2}{N}$$

Standard deviation of population

$$\sigma = \sqrt{\sum_{i=1}^{N} \frac{(x_i - \mu)^2}{N}}$$

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51

Variance and standard deviation of sample

Variance of sample

$$s^2 = \sum_{i=1}^{n} \frac{(x_i - \overline{x})^2}{(n-1)}$$

Standard deviation of sample

$$s = \sqrt{\sum_{i=1}^{n} \frac{(x_i - \overline{x})^2}{(n-1)}}$$

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52

Sample v.s. population standard deviation

$$\sigma = \sqrt{\sum_{i=1}^{N} \frac{(x_i - \mu)^2}{N}} \qquad s = \sqrt{\sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{(n-1)}}$$

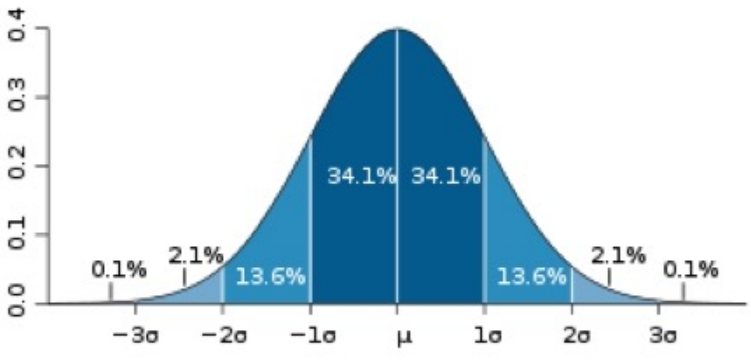
The reason for the change in formula with the sample is this: When you're calculating s you are normally using s^2 (the sample variance) to estimate σ^2 (the population variance). The problem, though, is that if you don't know σ you generally don't know the population mean μ, either, and so you have to use $\bar{x}$ in the place in the formula where you normally would use μ. Doing so introduces a slight bias into the calculation: Since $\bar{x}$ is calculated from the sample, the values of x_i are on average closer to $\bar{x}$ than they would be to μ, and so the sum of squares $\sum_{i=1}^{N}(x_i - \bar{x})^2$ turns out to be smaller on average than $\sum_{i=1}^{N}(x_i - \mu)^2$. It just so happens that that bias can be corrected by dividing by $N - 1$ instead of N. (Proving this is a standard exercise in an advanced undergraduate or beginning graduate course in statistical theory.) The technical term here is that s^2 (because of the division by $N - 1$) is an *unbiased estimator* of σ^2.

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Chebyshev theorem

For data with mean $\overline{x}$ and standard deviation *s:*
at least **75%** of the data will be in the interval of $\overline{x} \pm 2s$
and at least **89%** of the data will be in the interval of $\overline{x} \pm 3s$

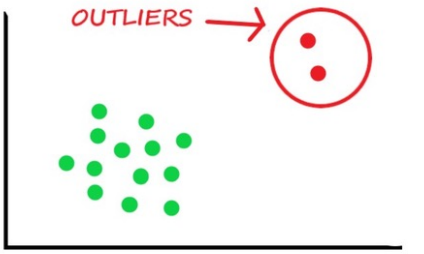


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54

Chebyshev theorem and outliers

- During data analysis, it is important to clean the data sample to ensure that the observations best represent the problem.
- Sometimes a dataset can **contain extreme values that are outside the range of what is expected** and unlike the other data.
- These are called outliers and often data analysis can be improved by understanding and even removing these outlier values.
- Outliers can have many causes, such as:
  - Measurement or input error.
  - Data corruption.
  - True outlier observation.



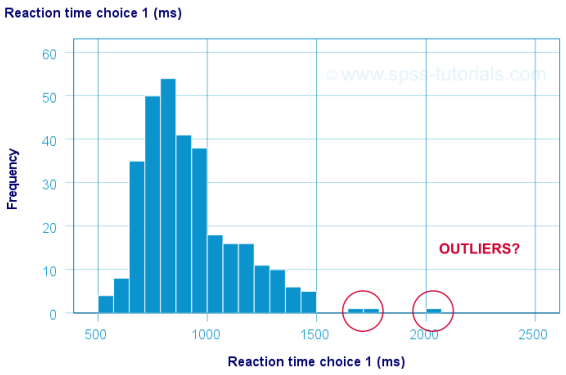
(Courtesy: Ou Zhang)

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Chebyshev theorem and outliers

- There is no precise way to define and identify outliers in general because of the specifics of each dataset.
- Instead, you, or a domain expert, must **interpret the raw observations** and decide whether a value is an outlier or not.
- Otherwise, we can use **statistical methods** to identify observations that appear to be rare or unlikely given the available data.

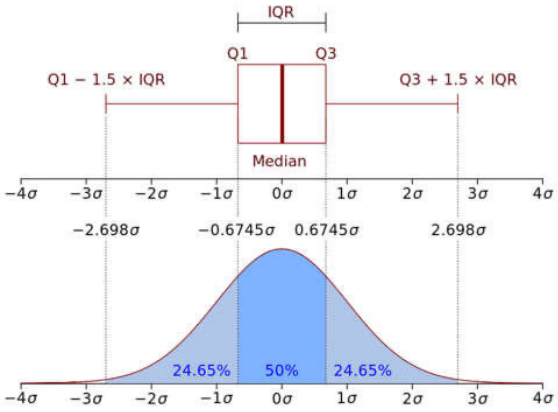


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Interquartile Range method

- The IQR is calculated as the difference between the 75th and the 25th percentiles of the data.
- Percentiles in our case is referred as quartiles (quart meaning 4) because the data is divided into four groups via the 25th, 50th and 75th values.
- The IQR can be used to identify outliers by defining limits on the sample values that are **a factor *k* of the IQR below the 25th percentile or above the 75th percentile.**
- The common value for the factor *k* is the value 1.5. A factor *k* of 3 or more can be used to identify values that are extreme outliers



Courtesy: Perez and Tah

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IQR in Python

- We can calculate the percentiles of a dataset using the **percentile()** NumPy function that takes the dataset and specification of the desired percentile.
- The IQR can then be calculated as the difference between the 75th and 25th percentiles.
- We can then calculate the cutoff for outliers as 1.5 times the IQR and subtract this cut-off from the 25th percentile and add it to the 75th percentile to give the actual limits on the data.
- We can then use these limits to identify the outlier values.

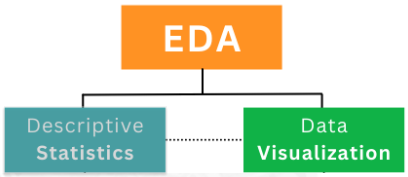
```
# identify outliers with interquartile range
from numpy.random import seed
from numpy.random import randn
from numpy import percentile
# seed the random number generator
seed(1)
# generate univariate observations
data = 5 * randn(10000) + 50
# calculate interquartile range
q25, q75 = percentile(data, 25), percentile(data, 75)
iqr = q75 - q25
print('Percentiles: 25th=%.3f, 75th=%.3f, IQR=%.3f' % (q25, q75, iqr))
# calculate the outlier cutoff
cut_off = iqr * 1.5
lower, upper = q25 - cut_off, q75 + cut_off
# identify outliers
outliers = [x for x in data if x < lower or x > upper]
print('Identified outliers: %d' % len(outliers))
# remove outliers
outliers_removed = [x for x in data if x >= lower and x <= upper]
print('Non-outlier observations: %d' % len(outliers_removed))
```

```
Percentiles: 25th=46.685, 75th=53.359, IQR=6.674
Identified outliers: 81
Non-outlier observations: 9919
```

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58

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General tips on data visualization

- Label the axes and unit
  - Does the scale of 0 to 100 percent represent task completion time? May not be obvious to audience
  - What are the time units?
  - If axis already makes it clear (e.g. "Task1", "Task2", etc.) don't need to add another label "Tasks"
- Don't imply more precision in your data then they deserve
- Don't use color alone to convey information
  - Color is a good principle for designing information display
  - What is the report is in black and white? Someone is color blind?
- Display labels horizontally whenever possible
  - Exception – title for vertical axis is normall vertical
- Show confidence intervals whenever possible
  - Mainly applies to bar graphs and line graphs that are presenting means of individual participant data
- Don't overload your graphs
  - Just because you can create a single graph that shows all of the ratings for 20 different tasks doesn't mean you should
- Be careful with 3D graphs
  - Does it really help? In some cases it's harder to see the values being plotted

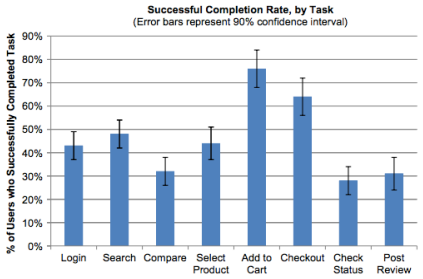
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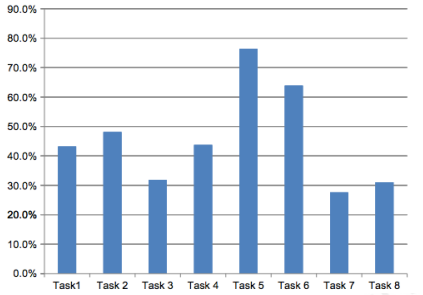
Presenting your data graphically

- Column or Bar Graphs
- Appropriate to present values of continuous data for discrete items or categories
  - If both continuous, use line graph
- Axis for continuous variable should start a zero
  - Length of bar indicates values
  - Don't artificially manipulate lengths
- Don't let maximum value for continuous variable go higher than it is theoretically able to
  - 110% percent

Good example



Bad example

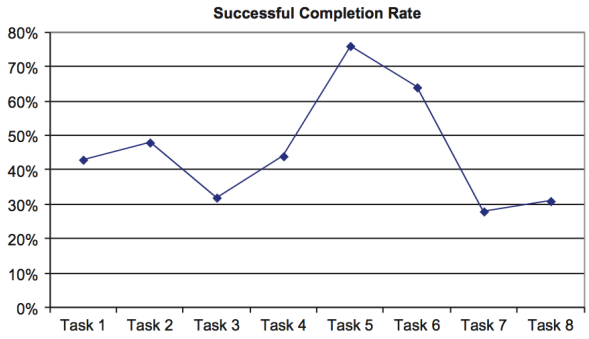


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Presenting your data graphically

- Line versus bar graph
  - In using a line graph ask if the places along the line between data points make sense
  - If not, use a bar graph



Question: should we present "task" as continuous data point? or discrete data point?

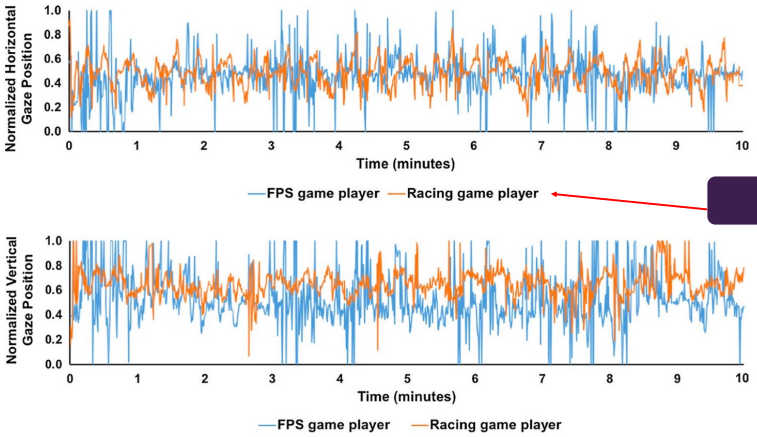
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62

Presenting your data graphically

Fig. 8 Samples of gaze positions during playing and watching the FPS and the racing games. The vertical and the horizontal axis denote gaze position and duration, respectively. Screen coordinates were normalized, so that $(x = 0, y = 0)$ and $(x = 1, y = 1)$ refers to the top left and the bottom right of the screen, respectively. The graphs show that the FPS game induced more intense vertical and horizontal saccadic eye movements compared with the racing game

You need to tell the story right away



(a) Sample of gaze positions in horizontal and vertical directions from participants playing FPS and racing game. Blue and orange line denote gaze position of FPS and racing game player, respectively

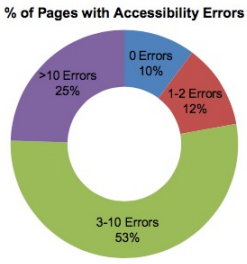
S. Wibirama, P. I. Santosa, P. Widyarani, N. Brilianto, W. Hafidh, "Physical Discomfort and Eye Movements during Arbitrary and Optical Flow-Like Motions in Stereo 3D Contents", Virtual Reality, Vol. 24, 2020, pp. 39-51.

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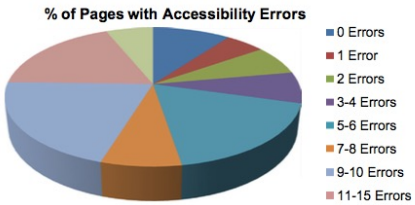
63

Presenting your data graphically

- Pie Charts
  - Illustrate parts or percentages of a whole
- Appropriate when the parts add up to 100 percent
  - To account for all cases you can add an "other" category
- Minimize number of segments
  - Difficult to make sense
- Include percentage and label for each segment



Good example



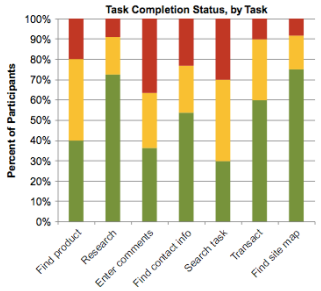
Bad example

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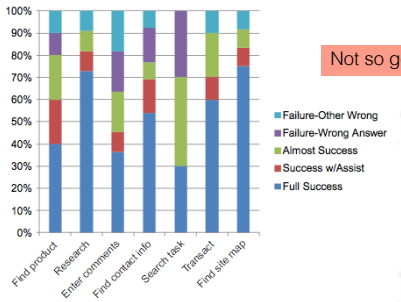
64

Presenting your data graphically

- Stacked Bar Graphs
  - Basically multiple pie charts shown in bar form
- Appropriate when parts for each item in a series add up to 100 percent
- Items in series are normally categorical
- Minimize number of segments in each bar
  - More than three can be difficult to interpret (except you write the percentage inside the bar)
- Make use of color coding conventions that audience is likely to be familiar with
  - In US – green is good, yellow is marginal, and red is bad



Good example



Not so good example

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65

Stem and leaf data presentation

The quantity of glucose in blood of 100 persons is measured and recorded in Table 2.0b (unit is mg%).

Table 2.0b *Quantity of Glucose in Blood of 100 Students (unit: mg%)*

70	79	80	83	85	85	85	85	86	86
86	87	87	88	89	90	91	91	92	92
93	93	93	93	94	94	94	94	94	94
95	95	96	96	96	96	96	97	97	97
97	97	98	98	98	98	98	98	100	100
101	101	101	101	101	101	102	102	102	103
103	103	103	104	104	104	105	106	106	106
106	106	106	106	106	106	106	107	107	107
107	108	110	111	111	111	111	111	112	112
112	115	116	116	116	116	119	121	121	126

Note: sort the data to facilitate stem-and-leaf creation

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Stem and leaf data presentation

GLUCOSE
GLUCOSE Stem-and-Leaf Plot

Frequency		Stem &	Leaf
1.00	Extremes		(=<70)
1.00		7 .	9
2.00		8 .	03
11.00		8 .	55556667789
15.00		9 .	011223333444444
18.00		9 .	556666777777888888
18.00		10 .	001111112223333444
16.00		10 .	5666666666677778
9.00		11 .	011111222
6.00		11 .	566669
2.00		12 .	11
1.00	Extremes		(>=126)

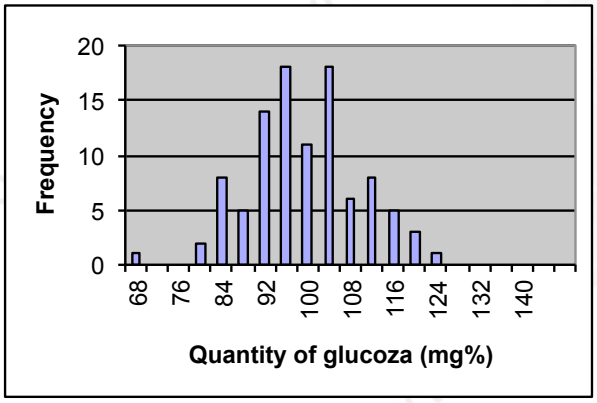
Stem width: 10
Each leaf: 1 case(s)

Stem width

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Histogram data presentation



The quantity of glucose in blood of 100 persons is measured and recorded in Table 2.0b (unit is mg%).

Table 2.0b *Quantity of Glucose in Blood of 100 Students (unit: mg%)*

70	79	80	83	85	85	85	85	86	86
86	87	87	88	89	90	91	91	92	92
93	93	93	93	94	94	94	94	94	94
95	95	96	96	96	96	96	97	97	97
97	97	98	98	98	98	98	98	100	100
101	101	101	101	101	101	102	102	102	103
103	103	103	104	104	104	105	106	106	106
106	106	106	106	106	106	106	107	107	107
107	108	110	111	111	111	111	111	112	112
112	115	116	116	116	116	119	121	121	126

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68

Take home messages

- Statistics is very important because we live in the era of data. Statistics is a science to learning information from data.
- We use statistics to model uncertainty. This model may be used to support decision making as well as prediction for future event.
- Variability is inevitable, especially in engineering experiment. We need to understand characteristics of the data, the distribution of data, how to visualize and analyze them.
- Descriptive statistics is used to understand the data. Inferential statistics is used to make prediction or generalization of population based on limited sample size.
- Errors are caused by many factors, one of them is outliers. Understanding statistics helps us to remove unwanted outliers.
- Data visualization is important to observe pattern in the data visually that might not be immediately apparent from the numerical summaries.



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69

Thank You

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70